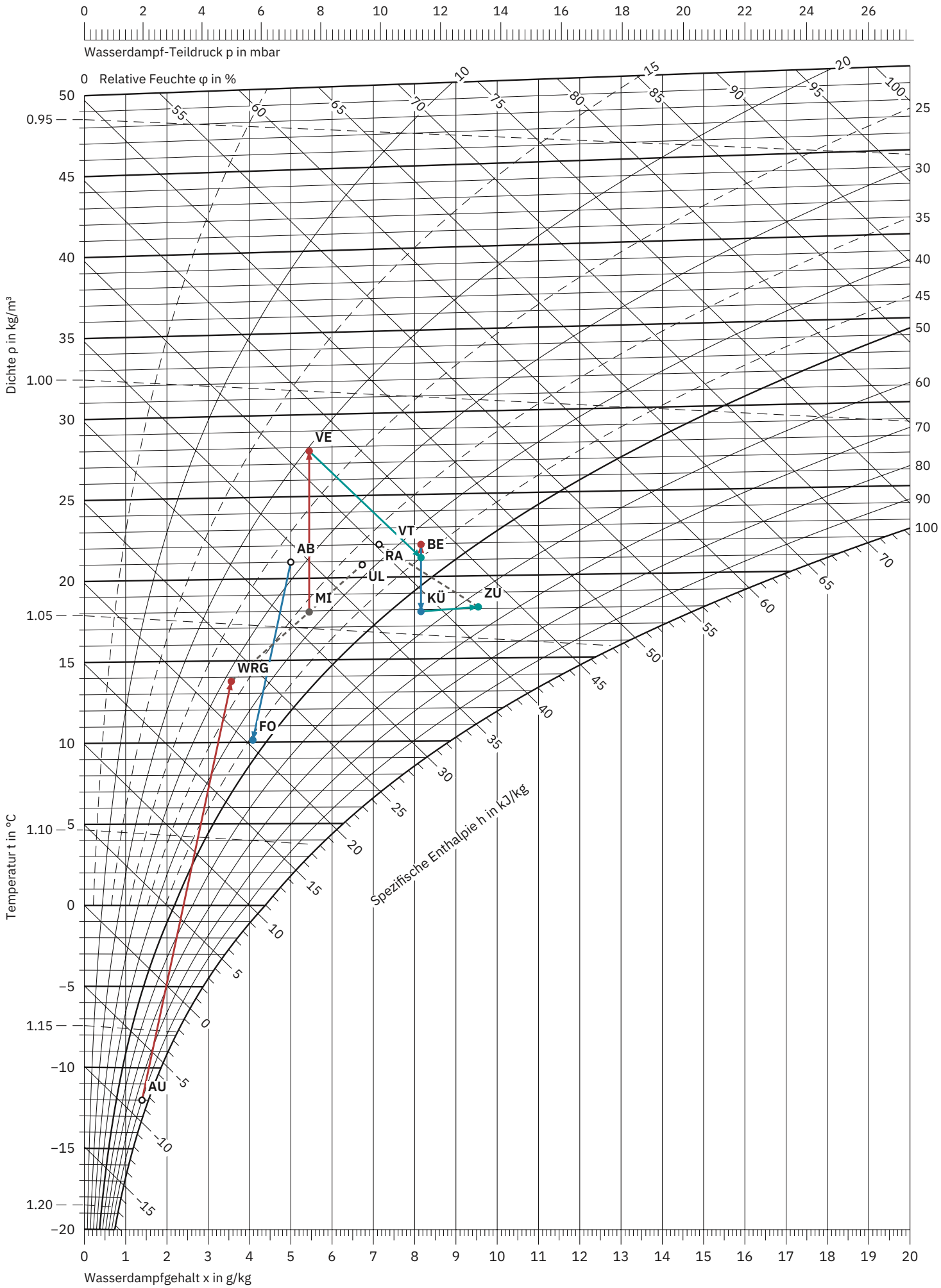
# h,x-Diagramm

p = 877 mbar, H = 1'200 m ü. M.

Objekt Alle Prozesse, 1200 m | Anlage Prozesse | Datum 21.10.2026



# h,x-Diagramm: Prozesse

| Projekt | Projekt-Nr. | Bearbeiter | Datum |
|---|---|---|---|
| Alle Prozesse, 1200 m | FX-10 | Fixture | 21.10.2026 |
| Luftdruck | V Berechnung | | |
| 877 mbar | 4'000 m³/h | | |

## Zustände

| Punkt | t °C | φ % | x g/kg | h kJ/kg | ρ kg/m³ | Taupunkt °C | Feuchtkugel °C | V m³/h | m kg/h |
|---|---|---|---|---|---|---|---|---|---|
| **AU** | -12.0 | 90.0 | 1.40 | -8.6 | 1.169 | -13.2 | -12.3 | 1'500 | 1'751 |
| **AB** | 21.0 | 28.0 | 5.00 | 33.8 | 1.036 | 1.8 | 10.6 | 4'000 | 4'122 |
| **WRG** | 13.7 | 31.6 | 3.56 | 22.8 | 1.063 | -2.5 | 6.0 | 1'654 | 1'751 |
| **FO** | 10.1 | 46.0 | 4.08 | 20.5 | 1.076 | -0.9 | 4.9 | 3'847 | 4'122 |
| **UL** | 20.8 | 38.1 | 6.73 | 38.0 | 1.035 | 6.0 | 12.2 | 2'500 | 2'571 |
| **MI** | 17.9 | 36.9 | 5.45 | 31.8 | 1.046 | 3.0 | 9.8 | 4'154 | 4'323 |
| **VE** | 27.8 | 20.3 | 5.45 | 41.8 | 1.012 | 3.0 | 13.6 | 4'294 | 4'323 |
| **BE** | 21.2 | 45.0 | 8.15 | 42.0 | 1.033 | 8.8 | 13.6 | 4'218 | 4'323 |
| **VT** | 22.0 | 42.8 | 8.15 | 42.8 | 1.030 | 8.8 | 13.9 | 4'229 | 4'323 |
| **KÜ** | 17.9 | 55.2 | 8.15 | 38.6 | 1.045 | 8.8 | 12.4 | 4'171 | 4'323 |
| **ZU** | 18.1 | 63.5 | 9.54 | 42.4 | 1.043 | 11.1 | 13.7 | 4'183 | 4'323 |
| **RA** | 22.0 | 37.5 | 7.14 | 40.3 | 1.031 | 6.8 | 13.0 | 4'000 | 4'094 |

## Prozesse

| Baustein | Eintritt | Austritt | Q kW | Wasser kg/h | Kennwert |
|---|---|---|---|---|---|
| WRG | AU + AB | WRG + FO | 15.3 | – | Φt = 78 % \| Φx = 60 % \| tFO,min = 0.0 °C |
| Mischung | WRG + UL | MI | – | – | Anteil 41 % |
| Lufterhitzer | MI | VE | 12.0 | – | Q = 12.0 kW |
| Adiabatbefeuchter | VE | BE | 0.2 | 11.7 | η = 46 % |
| Ventilator | BE | VT | 1.0 | – | Δt = 0.8 K |
| Luftkühler | VT | KÜ | -5.0 | 0.0 | tO = 12.0 °C \| Qs = -5.0 kW \| trocken |
| Dampfbefeuchter | KÜ | ZU | 4.5 | 6.0 | tD = 110 °C |
| Verbindung | ZU + RA | | -2.6 | – | Δh/Δx = 884 kJ/kg \| Δh = -2.1 kJ/kg |